Yu **Tian**

RESEARCH FELLOW · MAX PLANCK INSTITUTES PKS | CBG

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Research Interest

Network Theory, Network Dynamics and Optimisation, Mathematical Statistics and Data Science.

Academic Positions

Max Planck Institute for Physics of Complex Systems | Cell Biology and Genetics *Dresden, Germany*

RESEARCH FELLOW *Oct. 2024 -*

- ELBE Postdoctoral Fellow, Center for Systems Biology Dresden (CSBD)

Nordita, Stockholm University and KTH Royal Institute of Technology *Stockholm, Sweden*

RESEARCH FELLOW *Oct. 2022 - Oct. 2024*

- Funding: Wallenberg Initiative on Networks and Quantum information (WINQ)
- PIs: Prof. Frank Wilczek (MIT; Nobel Laureate) and Prof. John Wettlaufer (Yale University)

Education

University of Oxford *Oxford, UK*

PH.D. IN MATHEMATICS *Sep. 2018 - Oct. 2022*

- EPSRC Centre for Doctoral Training for Industrially Focused Mathematical Modelling (InFoMM CDT)
- Thesis: Role Extraction, Dynamics, and Optimisation on Networks; supervised by Prof. Renaud Lambiotte (Oxford), and industrial collaborators Dr. Alisdair Wallis, Dr. Sebastian Lautz (Tesco)

Honors & Awards

2025	**Grantee**, Travel Grant from Aalto Science Institute	*Finland*
2024	**Grantee**, Travel Scholarship from Wenner-Gren Foundation	*Sweden*
2024	**Fellow**, Simons Foundation Fellowship from Isaac Newton Institute	*UK*
2018	**Grantee**, EPSRC InFoMM CDT Studentship (fully-funded PhD studentship, 2018 - 2022)	*UK*

Publications & Preprints

*co-first author

PREPRINTS

16. **Y. Tian**, E. Wiesler, M. Weber. *Ollivier's Ricci curvature on complex-weighted graphs*. Preprint, arxiv:2608.10132, 2026. (Submitted to SIAM J. Math. Data Sci.)
15. **Y. Tian**, M. Porter, L. Böttcher. *Generalizing Perron–Frobenius theory and eigenvector-based centralities to networks with complex edge weights*. Preprint, arxiv:2606.12026 , 2026. (Submitted to SIAM J. Matrix Anal. Appl.)
14. **Y. Tian**, C. Subramanya, C. Modes. *GEMINI: Generalized Ensnarlment Measure from Incomplete-linkage of Network-network Interactions*. Preprint, arxiv:2606.05153, 2026. (Submitted to PNAS)

PEER-REVIEWED PUBLICATION IN MATHEMATICS AND PHYSICS

13. A. Gallo, **Y. Tian**, R. Lambiotte, and T. Carletti. *Global synchronization in matrix-weighted networks*. Commun. Phys., 8, 512, 2025.
12. **Y. Tian**, S. Kojaku, H. Sayama, and R. Lambiotte. *Matrix-weighted networks for modeling multidimensional dynamics: Theoretical foundations and applications to network coherence*. Phys. Rev. Lett., 134, 237401, 2025.
11. **Y. Tian**, and E. Estrada. *Balance with memory in signed networks via Mittag-Leffler matrix functions*. SIAM J. Matrix Anal. Appl., 46, 2, 1460–1483, 2025.
10. S. Babul, **Y. Tian**, and R. Lambiotte. *Strong and weak random walks on signed networks*. npj Complexity, 2, 4, 2025.
9. **Y. Tian***, Z. Lubberts* and M. Weber. *Curvature-based clustering on graphs*. JMLR, 26 (52), 1 - 67, 2025.
8. **Y. Tian** and R. Lambiotte. *Structural balance and random walks on complex networks with complex weights*. SIAM J. Math. Data Sci., 6(2), 372 - 399, 2024.
7. **Y. Tian** and R. Lambiotte. *Spreading and structural balance on signed networks*. SIAM J. Appl. Dyn. Syst., 23(1), 50 - 80, 2024.
6. X. Meng, X. Hu, **Y. Tian**, G. Dong, R. Lambiotte, J. Gao, and S. Havlin. *Percolation theories for quantum networks*, Entropy, 25(11), 1564, 2023.
5. **Y. Tian**, Z. Lubberts and M. Weber. *Mixed-membership community detection via line graph curvature*. NeurIPS Symmetry and Geometry in Neural Representations, 2023.
4. **Y. Tian** and R. Lambiotte. *Unifying information propagation models on networks and influence maximization*. Phys. Rev. E, 106, 034316, 2022.

PEER-REVIEWED PUBLICATION IN INTERDISCIPLINARY RESEARCH

3. R. Wang, **Y. Tian**, P. Liò, and G. Bianconi. *Dirac-equation signal processing: physics boosts topological machine learning*. PNAS Nexus, 4, 5, pgaf139, 2025.
2. L. Gyllingberg*, **Y. Tian***, and D. Sumpter. *A minimal model of cognition based on oscillatory and reinforcement processes*. J. R. Soc. Interface, 22, rsif.2024.0402, 2025.
1. **Y. Tian**, S. Lautz, A. Wallis and R. Lambiotte. *Extracting complements and substitutes from sales data: a network perspective*. EPJ Data Sci., 10:45, 2021.

Selected Presentations

Invited Talks

- **Geometric Machine Learning Seminar at Harvard University**, Cambridge, US (Jul. 2026)
- **Tropical Mathematics and Machine Learning Seminar at Yonsei University** (Virtual), Seoul, South Korea (May 2026)
- **Society and Networks Seminar at Aalto University**, Aalto, Finland (Apr. 2026)
- **Topological Statistics, Data and Intelligence Workshop**, Sanya, China (Feb. 2026)
- **Seminar at Statistical Physics Youth Communications** (Virtual). (Oct. 2025)
- **Seminar at Technical University of Dresden**, Dresden, Germany (Oct. 2025)
- **Math and AI Meeting jointly at MPI for Mathematics in Sciences, ScaDS.AI, and MPI for Human Cognitive and Brain Sciences**, Leipzig, Germany (Oct. 2025)
- **Computational Science Seminar at Frankfurt School**, Frankfurt, Germany (Aug. 2025)
- **SIAM Conference on Mathematics of Data Science**, Atlanta, US. (Oct. 2024)
- **Isaac Newton Institute Satellite Programme on Hypergraphs: theory and applications**, London, UK. (July 2024)
- **NERDS (NEtwoRks, Data, and Society) Seminar, IT University of Denmark**, Copenhagen, Denmark. (Apr. 2024)
- **Seminar at Technical University of Denmark**, Copenhagen, Denmark. (Apr. 2024)
- **Seminar at Niels Bohr Institute**, Copenhagen, Denmark. (Apr. 2024)
- **Nordic Workshop on Statistical Physics: Biological, Complex and Non-equilibrium Systems**, Stockholm, Sweden. (Mar. 2024)
- **Seminar at Vermont Complex Systems Center** (Virtual), Vermont, US. (Mar. 2024)
- **Seminar at Binghamton Center of Complex Systems** (Virtual), New York, US. (Feb. 2024)
- **Applied and Interdisciplinary Seminar at University of Bath**, Bath, UK. (Feb. 2024)
- **NetPLACE Seminar** (Virtual). (Feb. 2024)
- **Seminar at Imperial College London**, London, UK. (Feb. 2024)
- **CASA Seminar at University College London**, London, UK. (Feb. 2024)
- **Seminar at Queen Mary, University of London**, London, UK. (Jan. 2024)
- **Seminar at Linköping University**, Linköping, Sweden. (Sep. 2023)
- **Seminar at Indiana University** (Virtual), Bloomington, US. (Sep. 2023)
- **Seminar at Northwestern University** (Virtual), Chicago, US. (Aug. 2023)
- **BrainNet workshop at KTH**, Stockholm, Sweden. (May, 2023)
- **SIAM Conference on Applications of Dynamical System**, Portland, US. (May 2023)
- **Seminar at Uppsala University**, Uppsala, Sweden. (May 2023)
- **Soft Matter Seminar at Nordita**, Stockholm, Sweden. (Oct. 2022)
- **Seminar at Technical University of Munich** (Virtual), Munich, Germany. (May 2022)
- **Seminar at Dartmouth College** (Virtual), Hanover, US. (Jan. 2022)
- **Oxford Networks Seminar** (Virtual), University of Oxford, UK. (May 2021)

Contributed Talks

- **EMBO Workshop on Building networks: engineering in vascular biology**, Barcelona, Spain (May 2026)
- **DPG (German Physical Society) Spring Meeting**, Dresden, Germany (Mar. 2026)
- **RTCNS (Recent Trends in Coupled Network Systems) Workshop**, Berlin, Germany (Jan. 2026)
- **Biophysics Internal Seminar at MPI PKS**, Dresden, Germany (Nov. 2025)
- **HOOC (Higher Order Opportunities and Challenges) Workshop**, Aachen, Germany (Aug. 2025)
- **Conference on Network Sciences**, Maastricht, Netherlands. (June 2025)
- **Conference on Network Sciences**, Quebéc City, Canada. (June 2024)
- **British NetSci Symposium**, London, UK. (May 2024)
- **WINQ Mini-Workshop at Nordita**, Stockholm, Sweden. (Feb. 2023)
- **The 4th IMA Conference on The Mathematical Challenges of Big Data**, University of Oxford, UK. (Sep. 2022)
- **SIAM Workshop on Network Science** (Virtual). (Sep. 2022)
- **Conference on Network Sciences (NetSci)** (Virtual), Shanghai, China. (July 2022)
- **Industrial Mathematics in the 21st Century: A cornucopia of unsolved problems**, University of Oxford, UK. (June 2022)
- **InFoMM Annual Meeting**, University of Oxford, UK. (June 2022)
- **SIAM UKIE National Student Chapter Conference**, Edinburgh, UK. (June 2022)
- **The 13th International Conference on Complex Networks (CompleNet)** (Virtual), Exeter, UK. (May - June 2022)
- **The 10th International Conference on Complex Networks and their Applications (CNA)**, Madrid, Spain. (Nov. 2021)
- **Conference on Complex Systems (CCS)**, Lyon, France. (Oct. 2021)
- **InFoMM Annual Meeting** (Virtual), University of Oxford, UK. (July 2021)

Teaching Experience

MPI CBG | Dresden International PhD Program — *Dresden, Germany*
Co-Lecturer — *Sep. 2025 - Nov. 2025*

- Physical Principles in Biology, 2025
- Topics: Dynamics Systems; Complex Systems and Networks

University of Oxford — *Oxford, UK*
Tutor — *Oct. 2021 - Dec. 2021*

- Introduction to Statistics, Michaelmas Term 2021

Teaching Assistant — *Oct. 2019 - Apr. 2020*

- Networks, Hilary Term 2020
- Graph Theory, Michaelmas Term 2019

Professional Experience

Reviewers

- **Journals**: Science Advances, Communication Physics, Scientific Reports, PNAS Nexus, Physical Review E, PLOS Complex Systems, Physica A: Statistical Mechanics and its Applications, Advances in Complex Systems, IEEE Transactions on Signal and Information Processing over Network, Pattern Recognition.
- **Conferences**: International AAAI Conference on Web and Social Media, NeurIPS New Perspectives in Graph Machine Learning.

Organizers

- **Discrete Laplacian Workshop**, MPI CBG — *Jun. 2026*
- **WINQ Seminar**, Nordita — *Oct. 2022 - Oct. 2024*
- **WINQ Workshop on Complex Dynamical Networks**, Nordita — *Apr. 2024 - May 2024*
- **Session at INFORMS 2023 Annual Meeting**, Phoenix — *Oct. 2023*
  Session Title: Networks in Machine Learning and Data Science
- **WINQ Mini-Workshop**, Nordita — *Feb. 2023*
- **Oxford Networks Seminar**, University of Oxford — *Oct. 2021 - June 2022*

Service & Outreach

- **Equity, Diversity and Inclusion (EDI) Committee** Member, Nordita — *Oct. 2023 - Oct. 2024*

Industrial Experience

University of Oxford — *Oxford, UK*

InFoMM CDT Mini-Project with *Air Products*: Inter-District Packaged Gas Optimisation — *Jul. - Sep. 2019*

- Formulated the problem of both inventory management and transshipment of products as a mixed integer programming.
- Proposed several relaxation methods based on Lagrangian relaxation to improve solving efficiency.

InFoMM CDT Mini-Project with *Tesco*: Halo Effect and Demand Transfer on Products — *Apr. - Jun. 2019*

- Devised a method combining Poisson processes with time series analysis to identify the product relationships from sales data.
- Applied regression techniques, and proposed several validation methods with real data.

Technology Skills

- **Programming:** Proficiency in Python (pandas, numpy, scipy, networkx, statsmodels, scikit-learn etc), MATLAB; Familiarity with R Language, C Language.
- **Optimisation:** MOSEK, Lingo.